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Structure refinement and diffuse streak scattering of silk (*Bombyx mori*)

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Abstract

Reexamination of the crystal structure of silk (*Bombyx mori*) was carried out by X-ray diffraction method. Four molecular chains are contained in the rectangular unit cell with parameters, $a = 9.38$ Å, $b = 9.49$ Å, and c (fiber axis) $= 6.98$ Å, and the space group $P2_1-C_2^2$. Silk assumes the statistical crystal structure, in which two antipolar–antiparallel sheet structures with different orientations statistically occupy a crystal site with the ratio 2:1. The molecular conformation is essentially the same pleated sheet structure as Marsh, Corey and Pauling proposed. However, the sheet structure formed by hydrogen bonds assumes the antipolar–antiparallel structure different from that proposed by Marsh, Corey and Pauling, in which the methyl groups of alanine residues alternately point to both sides of the sheet structure along the hydrogen bonding direction. The crystalline region of silk is composed of stacking of two antipolar–antiparallel sheet structures with different orientations. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Silk; Crystal structure; Structure refinement

1. Introduction

The X-ray diffraction studies of silk (*Bombyx mor*i) started from the work of Nishikawa and Ono in 1913 [1], in which they pointed out the characteristic feature of the so-called 'fiber diagram'. Thereafter, the structural studies of the silk fibroin have been made by many researchers [2–5]. In 1955, the crystal structure of silk fibroin, a regular arrangement of antiparallel sheets, was proposed by Marsh, Corey and Pauling [6], which has been accepted so far. However, their crystal structure model is based on the quantitative intensity estimation of six equatorial reflections (discrepancy factor $R = 37\%$) and the qualitative intensity estimation of the other 24 reflections on the equator and the layer lines. Judging from the present level of the crystal structure analysis of the fibrous sample, it is insufficient to be accepted. In the present study, the reexamination of the crystal structure of silk was carried out. Consequently, the crystalline region of silk was clarified to be of rather random stacking of antipolar-antiparallel sheets, in which the methyl groups (β-carbon) of alanine residues alternately point to both sides of the sheet structure along hydrogen-bonding direction, differing from the crystal structure model proposed by Marsh et al. [6]. In the previous papers [7,8], the crystal structure analysis was briefly reported. The purpose of the present paper is fully to describe the procedure of the structure analysis and to clarify the origin of the diffuse streak scatterings.

2. Experimental

The cocoon fibers were purified according to the following procedure: (a) degummed by refluxing in about 0.05% soapy water for over 2 h; (b) washed in 1% sodium carbonate solution to remove the remained soap; (c) rinsed in distilled water and in water acidified by acetic acid (pH 3) several times; and (d) washed in ether to remove wax and dried in vacuo. The fibers thus obtained were rolled up on a metal holder in a shape of

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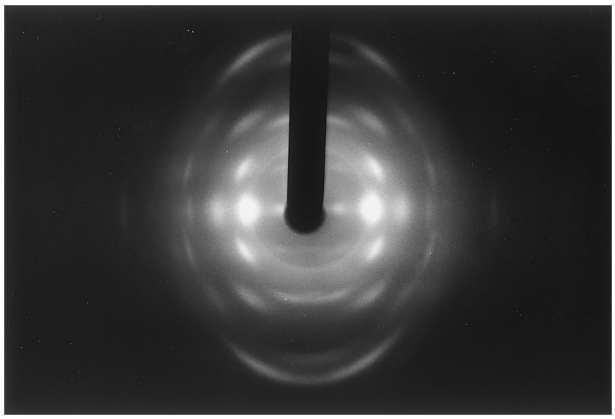


Fig. 1. Fiber diagram of silk.

cylinder with 1 mm diameter and served for taking the fiber diagram and PSPC measurements. The specimen used for Weissenberg photograph was prepared by cutting the sample off the cylinder after gluing the filaments together with cyanoacrylate. The doubly oriented sample was prepared by rolling the silk from the gland lumen.

X-ray measurements were carried out by CuKα radiation monochromatized by a pyrolyzed graphite. The cylindrical vacuum cameras with 5 and 10 cm radii and a camera with 4.5 cm radius were used for taking fiber diagrams and Weissenberg photographs, respectively. The fiber diagram of silk fibroin is shown in Fig. 1. The Weissenberg photograph was taken according to Norman's method, which is given in Fig. 2 along with its schematic representation. The photograph and schematic representation of a doubly oriented sample are shown in Fig. 3. The PSPC (position sensitive proportional counter) measurements were made by Ni-filtered CuKα radiation. The intensities of four strong reflections were measured by a PSPC system. The integrated intensities of 26 reflections observed on the fiber and Weissenberg photographs taken by the multiple film method were measured by using a drum scan densitometer (Optronics) installed in the Crystallographic Research Center of Osaka University. The optical density of a 100 μm-square was measured over the whole film and was digitally recorded on a magnetic tape. After the optical densities were converted to intensities in reference to a standard intensity scale, they were summed up along an arc or a line which had a constant diffraction angle 2θ. The summed intensities were plotted against a layer line for the fiber diagram and 2θ for the Weissenberg photograph. They are graphically integrated and finally corrected by the Lorentz-polarization factor and for obliquity. Five very weak reflections

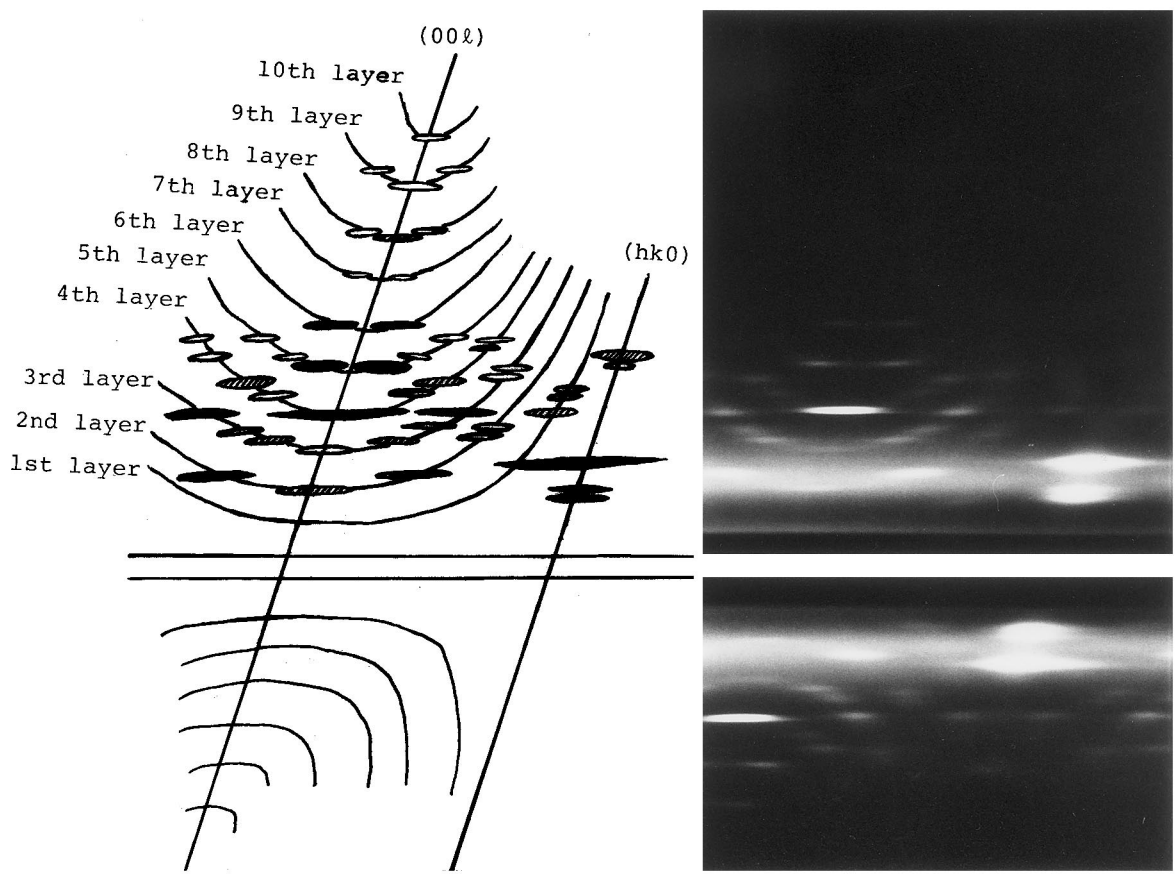


Fig. 2. Weissenberg photograph of silk and its schematic representation.

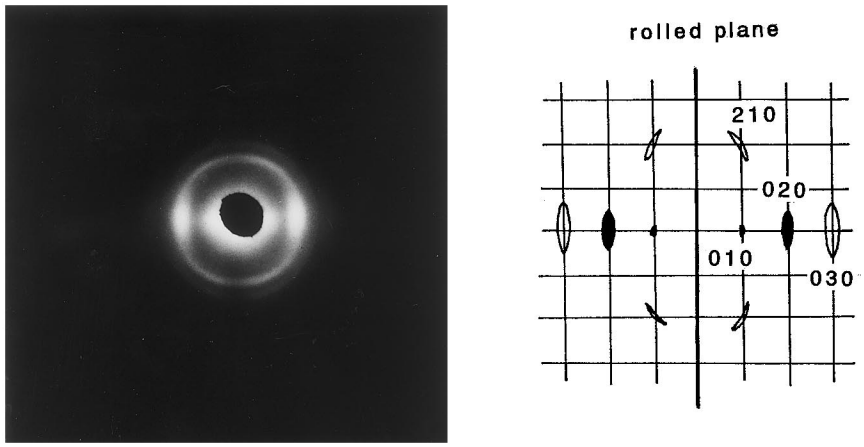


Fig. 3. X-ray diffraction photograph of the doubly oriented sample of silk and its schematic representation.

were visually estimated. The strongest reflection on the equator was separated into two components (020) + (200) and (120) + ($\bar{1}$20) + (210) + ($\bar{2}$10) by the least-squares method under the assumption of Gaussian profile. Thus, the intensities of 35 independent reflections were obtained.

3. Results and discussion

3.1. Fiber period and molecular model

The silk fibroin is a protein rich in glycine (43.7%), alanine (28.8%) and serine (11.9%) [9], and is composed of H-chain (molecular weight: 350 000) and L-chain (molecular weight: 25 000) linked by a disulfide bond [10]. Amino acid sequences of the L-chain and part of the H-chain were clarified [11–13]. From the amino acid sequence [11], the L-chain cannot be considered to be associated with the crystalline region of silk. The rather regular part of the H-chain, C_p fraction, which is obtained by treatment of the fibroin solution with an enzyme, chymotrypsin, is considered to be associated with the crystalline region [2]. The C_p fraction has the following amino acid sequence [13],

Y-G-A-G-A-G-[-S-G-(-A-G-)$_2$-]$_8$-S-G-A-A-G-Y

where G, A, S and Y denote glycine, alanine, serine and tyrosine, respectively. Accordingly, the amino acid sequence of silk fibroin in the crystalline region is considered as (-GAGAGS-)$_n$. As a first approximation, the amino acid sequence is considered to be (-GA-)$_n$. The fiber identity period was estimated as 6.98 Å from the fiber diagram, although the very weak streak layers corresponding to the sequence (-GAGAGS-)$_n$ are observed between the equator and the first layer line. From the fiber identity period, the molecular conformation is considered to be essentially the same pleated-sheet structure having $(TS\bar{S}TS\bar{S})_n$ conformation as Marsh et al. [6] proposed, where T is *trans*, S is skew, and $\bar{S}$ is minus skew conformations.

3.2. Unit cell and sheet structure models

All the observed reflections can be indexed by the rectangular unit cell with parameters, $a = 9.38$ Å, $b = 9.49$ Å, and c (fiber axis) $= 6.98$ Å and the space group $P2_1$–C_2^2, which are essentially the same as those reported by Marsh et al. [6]. The unit cell contains four molecular chains. The symmetries and four molecular sites in the unit cell with the space group $P2_1$ are shown in Fig. 4, along with the schematic representation of the molecular orientation. The molecules at site 1 (1′) and at site 2 (2′) are symmetrically independent of each other and form a sheet structure parallel to the *ac*-plane by hydrogen bonds. Two molecules at sites 1 and 2 can

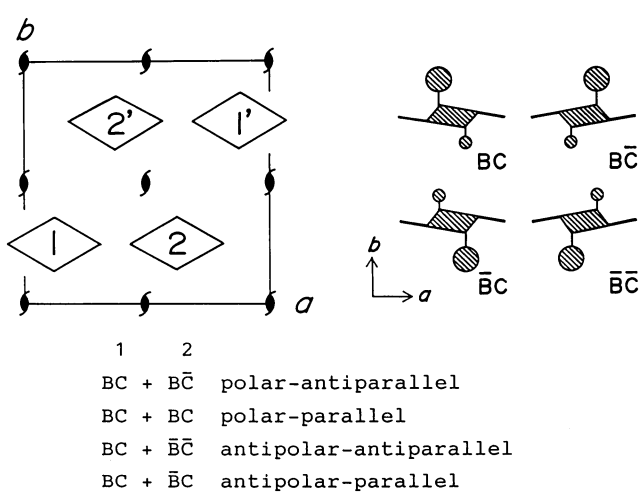


Fig. 4. The symmetries and four molecular sites in the unit cell with the space group $P2_1$ and the schematic representation for molecular orientation of silk.

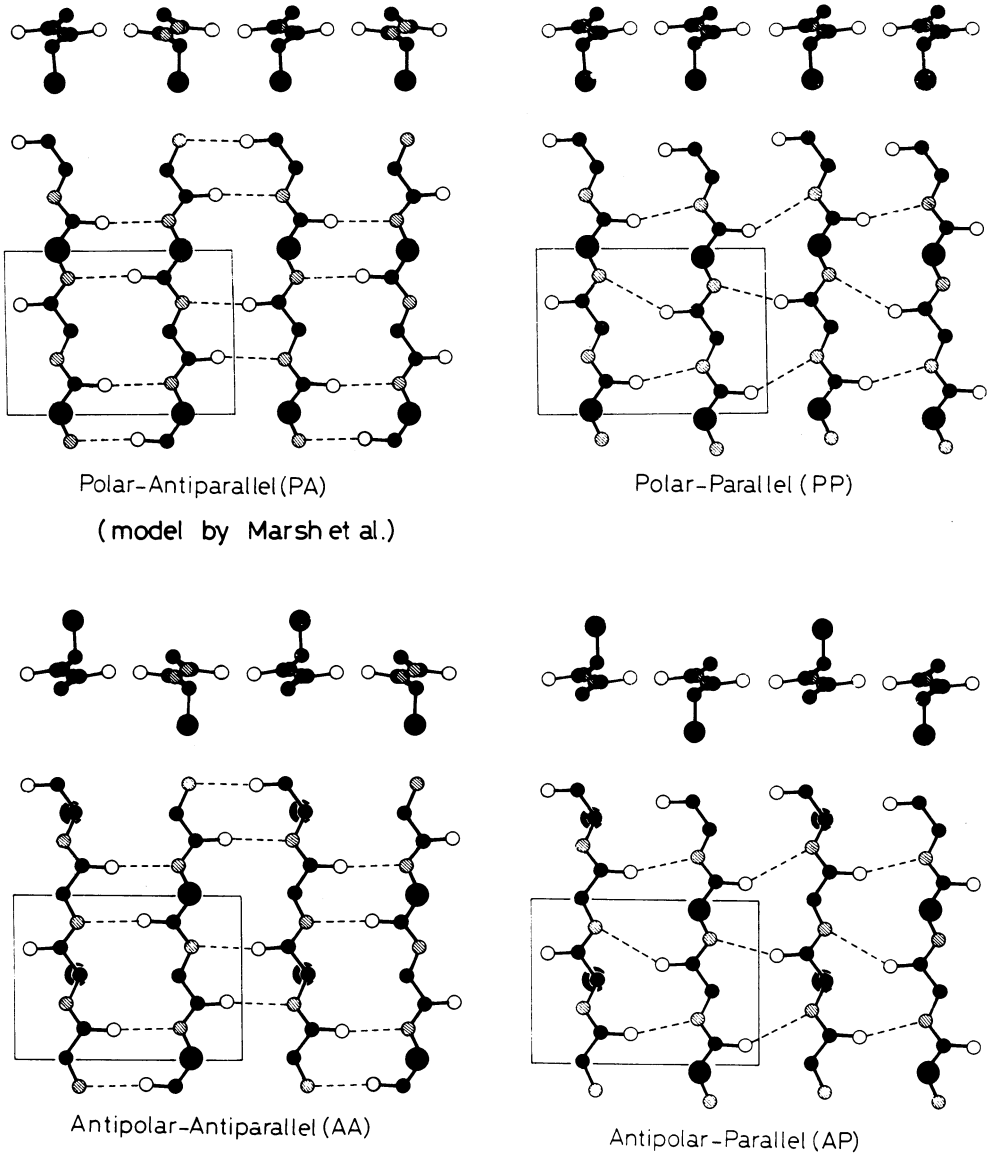


Fig. 5. Four models for the sheet structure formed by hydrogen bonds.

be related to the molecules at sites 1′ and 2′ by the 2-fold screw symmetry, respectively. Four models can be built up for the hydrogen bonding sheet, polar–antiparallel sheet (PA), polar–parallel sheet (PP), antipolar–antiparallel sheet (AA), and antipolar–parallel sheet (AP) models (Fig. 5). The molecules in the parallel sheet model assume the same direction, while in the antiparallel sheet model, the up- and down-molecules alternately arrange along the sheet. In the polar model, the methyl groups of alanine residues are only on one side of the sheet, while in the antipolar model, the methyl groups alternately point to both sides of the sheet along the hydrogen bonding direction. The crystal structure proposed by Marsh et al. corresponds to the regular arrangement of the polar–antiparallel (PA) sheets [6].

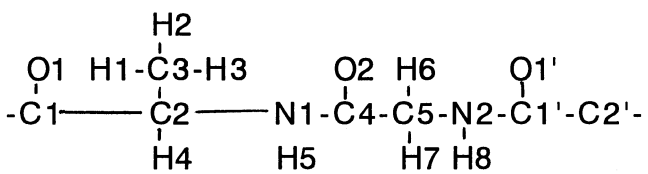
3.3. Refinement under regular arrangement of sheets

The constrained least-squares method [14,15] was applied to the refinement of the silk after it was slightly modified (see Appendix). During the refinements, the bond lengths and angles were fixed on the standard values which were adopted in reference to the values of some low molecular weight model compounds [16–20]. The adopted values for bond lengths and bond angles are listed in the Table 1. The same conformation was assumed for the symmetrically independent molecules. The molecule can assume four different orientations, BC, $B\bar{C}$, $\bar{B}C$ and $\bar{B}\bar{C}$ at a crystal site (Fig. 4). The polar–antiparallel sheet model (PA model) can be constructed by placing the molecules with orientation BC at crystal site 1 and with orientation $B\bar{C}$ at crystal site

Table 1
Bond lengths and bond angles adopted for silk[a]

Bond lengths (Å)		Bond angles (°)	
C1–O1	1.225	O1–C1–C2	121.0
C4–O2	1.225	O2–C4–C5	121.0
C1–C2	1.525	C2–N1–H5	116.1
C4–C5	1.525	C5–N2–H8	116.1
C2–N1	1.45	C2–N1–C4	122.5
C5–N2	1.45	C5–N2–C1′	122.5
N1–C4	1.335	C4–N1–H5	123.0
N2–C1′	1.335	C1′–N2–H8	123.0
C2–C3	1.525	N1–C4–C5	114.5
		N2–C1′–C2′	114.5
C–H	1.07	N1–C4–O2	124.5
N–H	1.00	N2–C1′O1′	124.5
		Others	109.5

[a] Numbering of atoms is given as follows,



2. Two molecules in the sheet are related by the 2-fold rotation axis parallel to the *b*-axis. The polar–parallel sheet model (PP model) can be constructed by placing the molecules with the same orientation at crystal sites 1 and 2. Two molecules in the sheet are related by the simple translation. The antipolar–antiparallel sheet model (AA model) can be constructed by placing the molecules with orientation *BC* at crystal site 1 and with orientation $\bar{B}\bar{C}$ at crystal site 2. Two molecules in the sheet are related by the 2-fold rotation axis parallel to the *a*-axis. The antipolar-parallel sheet model can be constructed by placing the molecules with orientation *BC* at crystal site 1 and with orientation $\bar{B}C$ at crystal site 2. Two molecules in the sheet are related by the 2-fold rotation axis parallel to the *c*-axis. The positions and the azimuthal angles of the two molecules at crystal sites 1 and 2 are independently refined. Furthermore, the following anisotropic temperature factor was taken into account,

$$\exp[-\{(ha*/2)^2 B_x + (kb*/2)^2 B_y + (lc*/2)^2 B_z\}]$$

where B_x, B_y and B_z are the contribution of the vibration and disorder in the *a*-direction along the hydrogen bonds, in the *b*-direction perpendicular to the sheet, and in the *c*-direction along the molecular chain, respectively. The refinement was first made for the models based on the regular arrangement of the sheets. The parameters to be refined were the scale factor, the Cartesian coordinates (X_0, Y_0) of the oxygen atom chosen as the origin of the molecule at site 1, Euler angles (θ, ϕ, χ), the fractional coordinates (x_2, y_2, z_2) for the translation of the molecule to site 2, the azimuthal angle of the molecule at site 2, five internal rotation angles, O1–C1–C2–N1, C1–C2–N1–C4, C2–N1–C4–C5, N1–C4–C5–N2 and C4–C5–N2–H8, and three anisotropic temperature parameters, B_x, B_y and B_z. Numbering of the atoms is given at the footnote of Table 1. Lagrange's undetermined multipliers are used for constraining the distances between the neighboring units in a molecular chain, N2…C1′, C5…C1′, H8…C1′, N2…C2′, and N2…O1′.

Finally, *R*-factors reduced to 17.9% for the polar-antiparallel model (PA-I), 19.8% for the polar–parallel model (PP-I), 24.5% for the antipolar–antiparallel model (AA-I), and 18.3% for the antipolar–parallel model (AP-I) (Table 2). Hereafter, *R*-factors are estimated only for the observed reflections. The PA-I model, which corresponds to the structure proposed by Marsh et al. [6] gives the lowest *R*-factor at this stage, but these values are not necessarily sufficient to distinguish which model is correct and furthermore does not give the stereochemically reasonable hydrogen bonding networks: hydrogen bond distances are different and asymmetric on both sides of the molecule.

Table 2
Structure models of silk and *R*-factors

Models	Probabilities (%)								
	BC	$B\bar{C}$	$\bar{B}C$	$\bar{B}\bar{C}$	NF[a]	B_x	B_y	B_z	R
PA-I	100	–	–	–	13	4.5	37.8	10.3	17.9
PA-II	22	64	14	1	15	2.5	28.0	5.2	13.7
PP-I	100	–	–	–	13	6.2	33.8	10.4	19.8
PP-II	36	51	9	4	15	2.2	21.4	4.5	13.8
AA-I	100	–	–	–	13	3.3	28.6	12.8	24.5
AA-II	25	70	2	3	16	2.8	34.3	9.5	9.7
AP-I	100	–	–	–	13	4.4	33.5	9.5	18.3
AP-II	28	61	1	10	16	2.5	15.1	6.1	11.6

[a] Number of freedoms.

3.4. Refinement under disordered arrangement of sheets

Many researchers pointed out the inter- and intra-sheet disorder on the silk and related polypeptides [21–23]. The anisotropy in the reflection broadening observed on X-ray diffraction pattern, especially of the doubly oriented sample (Fig. 3), suggests that the silk includes the disorder in the direction perpendicular to the hydrogen bonding sheets, i.e. the stacking disorder of the hydrogen bonding sheets (intersheet disorder). The disorder can be taken into account;the crystal structure models. The disorder reduces to the statistical structure as an averaged structure, in which four sheets with different orientations statistically coexist at a crystal site with different probabilities (models PA-II, PP-II, AA-II, AP-II in Table 2). These models are actually constructed by allotting the different probabilities to the molecules with different orientations at site 1, BC, $B\bar{C}$, $\bar{B}C$ and $\bar{B}\bar{C}$ (Fig. 4). This type of disorder was found in the crystal structures of poly(vinylidene fluoride) form II and it-polypropylene [24–26]. Four sheets statistically coexisting at a crystal site are created by three kinds of 2-fold rotation axes: 2-fold rotation axes parallel to the a-axis at the point $y = 0.25$ and $z = 0.375$, parallel to the b-axis at the point $x = 0.125$ and $z = 0.375$, and parallel to the c-axis at the point $x = 0.125$ and $y = 0.25$. The three statistical weights (existence probabilities) of the molecules, W_{BC}, $W_{\bar{B}C}$ and $W_{B\bar{C}}$ and the z-coordinate of the origin atom of the molecule, Z_0, were introduced to the parameters to be refined in addition to those used in the regular arrangement of the sheet structures. Here, the fourth statistical weight is given by the following equation,

$$W_{\bar{B}\bar{C}} = 1 - W_{BC} - W_{\bar{B}C} - W_{B\bar{C}}$$

The refinement gave far better agreement between the observed and calculated structure factors than the models with regular arrangement of the sheet structures. R-factors reduced to 13.7% for the polar–antiparallel model (PA-II), 13.8% for the polar–parallel model (PP-II), 9.7% for the antipolar–antiparallel model (AA-II), and 11.6% for the antipolar–parallel model (AP-II). The results are summarized in Table 2 along with PA-I, PP-I, AA-I, and AP-I models. The AA-II model gave the lowest R-value 9.7%, although, in the regular arrangement, the AA model gave the largest R-value 24.5% (AA-I model). This suggests that the AA-II model is more rigorous.

Subsequently, the serine residue was taken into consideration by replacing the 1/3 of alanine by the 1/3 of serine in weight. Refinements were made for the models that the C(=O)–C(α)–C(β)–O(H) assumes, T, G and $\bar{G}$ and assumes the statistical combinations of T and G, of T and $\bar{G}$, of G and $\bar{G}$ and of T, G and $\bar{G}$ (Table 3) where G and $\bar{G}$ denote gauche and minus gauche conformations. The R-factor did not reduce in the cases of the polar–antiparallel (PA-II) and the polar–parallel (PP-II) models. However, the antipolar–antiparallel (AA-II) model gave the lowest R-value 8.5% when the internal rotation angle C(=O)–C(α)–C(β)–O(H) assumes gauche conformation. From the above-mentioned facts, it can be concluded that the AA-III model is most probable for the crystal structure of silk fibroin. In the model AA-III with the gauche conformation of C(=O)–(α)–C(β)–O(H), the probabilities for the sheets with different orientations occupying a crystal site are 72, 24, 1 and 4%. Accordingly, within the accuracy of the present study, it can be said that a crystal site is statistically occupied by two antipolar–antiparallel sheets (AA) related by a 2-fold rotation axis parallel to the b-axis.

Table 3
Introduction of serine

Conformation C–C(α)–C(β)—O of serine	R-factors (%)			
	PA-II	PP-II	AA-II	AP-II
T	17.0	16.3	11.5	12.5
G	14.0	13.9	8.5	11.7
$\bar{G}$	14.3	13.3	12.1	10.7
TG	14.2	13.3	12.1	10.7
$T\bar{G}$	14.2	13.8	11.1	12.5
$TG\bar{G}$	13.9	14.0	10.1	11.3
$(\text{-GA-})_n$	13.7	13.8	9.7	11.6

3.5. Final refinement of the crystal structure

Finally, the refinement was carried out for the crystal structure in which two antipolar–antiparallel sheets related by a 2-fold rotation axis parallel to the b-axis statistically occupy a crystal site. R-factor reduced to 7.5%. The refined parameters and the final values are summarized in Table 4. Here, the azimuthal angles of the two molecules forming a sheet were not refined independently because the R-factor was not improved by taking them into account. The internal rotation angle C(=O)–C(α)–C(β)–O(H) of serine residue was refined. Table 5 gives the fractional coordinates. Table 6 gives the comparison between the observed and calculated structure factors. Some non-observed reflections give rather large values for the calculated structure factors. Here, it should be noted that the threshold for the non-observed reflections is determined by the reflection broadening and background intensity in addition to the calculated structure factor. The crystal structure is shown in Fig. 6.

3.6. Crystal structure and disorder of silk

In Fig. 7, the molecular structure of silk is shown in comparison with the molecular model proposed by

Marsh et al. [6] The molecular structure is essentially the same pleated sheet structure as Marsh et al. proposed. The internal rotation angles ϖ of the glycine and alanine residues about N–C(=O) bonds are 164.5° and 164.5°, respectively, which are nearly *trans* conformation. The internal rotation angles ϕ and ψ for the glycine residue are $-140.2°$ and 153.2°, respectively, which are between skew and *trans* conformations. The internal rotation angles ϕ and ψ for the alanine residue are $-141.1°$ and 152.2°, respectively, which are also between skew and trans conformations. These values can be compared with $\phi = -149.9°$, $\psi = 146.5°$ and $\varpi = 180°$ of polyglycine I [27] and $\phi = -139°$, $\psi = 135°$, and $\varpi = -178°$ of β-poly(L-alanine) [21]. In the final structure, the internal rotation angle C(=O)–C(α)–C(β)–O(H) is 37.6° and C(β)–O(H) bond is almost along the chain direction.

The hydrogen bonding network is shown in Fig. 8. In the network, the hydrogen bonds between O atoms of the glycine and N atoms of alanine are 2.76 and 3.05 Å, which can be compared with 2.83 Å of poly(L-alanine) with antiparallel pleated sheet structure. The hydrogen bonds between O atoms of the alanine and N atoms of the glycine are a little longer, 3.17 and 3.41 Å. The intermolecular distances between O(H) atom of serine residue and O(=C) atom of alanine residue are 3.05 and 3.35 Å, which may suggest that the OH groups of the serine residues are associated with the hydrogen bonding network forming the sheet structure.

The intermolecular distances shorter than the sums of van der Waals radii are shown in Fig. 9. Both of them are associated with the O(H) atom of serine residue. The shortest contact 2.59 Å is the distance between the atoms O(H) of serine residues with the same height. This suggests that when the sheet structures stacks, the serine residues are not located at the same height or that the hydrogen bonds are formed between the sheet structures. The distance 2.90 Å may suggest that the sheet structures is slightly distorted by the existence of serine residues.

Silk assumes the statistical crystal structure in which two antipolar–antiparallel sheet structures with different orientations statistically occupy a crystal site. This statistical structure is brought about by the disorder in stacking of two kinds of antipolar–antiparallel sheet structures with different orientations. This is consistent with the facts that the reflections 0k0 are observed to be broad on the X-ray diffraction pattern of the doubly

Table 4
Final parameters obtained by the constrained least-squares refinement

	Values	S.D.
Cartesian coordinates of the origin atom O (Å)		
X_O	3.28	0.06
Y_O	1.37	0.09
Z_O	1.35	0.07
Eulerian angles (°)		
θ	61.2	0.3
ϕ	−102.0	3.9
χ	−88.5	1.4
Molecular shift to site 2 in fractional coordinates		
x_2	0.515	0.010
y_2	0.483	0.004
z_2	1.326	0.017
Probability		
W_{BC}	0.32	0.04
Anisotropic temperature parameter (Å²)		
B_x	4.98	0.72
B_y	37.0	4.95
B_z	9.43	1.06
Internal rotation angles (°)[a]		
O1–C1–C2–N1	−40.3	3.4
C1–C2–N1–C4	−143.2	3.7
C2–N1–C4–C5	164.1	2.7
N1–C4–C5–N2	154.4	2.6
C4–C5–N2–H8	55.8	3.2
C1–C2–C3–O(H)[b]	37.6	9.3

[a] Numbering of atoms is given at the footnote of Table 1.
[b] O(H) denotes the oxygen atom of serine residue.

Table 5
Fractional coordinates of the atoms[a]

	x	y	z
O1	0.350	0.144	0.194
C1	0.222	0.169	0.190
C2	0.150	0.251	0.351
C3	0.178	0.408	0.325
O(H)	0.173	0.442	0.126
N1	0.207	0.205	0.534
C4	0.127	0.194	0.691
O2	0.009	0.247	0.710
C5	0.195	0.108	0.852
N2	0.136	0.153	1.035
H1	0.174	0.433	0.176
H2	0.097	0.467	0.399
H3	0.280	0.434	0.382
H4	0.038	0.231	0.348
H5	0.312	0.183	0.536
H6	0.173	−0.001	0.831
H7	0.308	0.125	0.852
H8	0.030	0.149	1.047

[a] Numbering of the atoms is given at the footnote of Table 1. Fractional coordinates for the molecule forming sheet structure are given by the following equation,

$$\mathbf{x}' = \mathbf{P}\mathbf{x} + \mathbf{x}_2 \qquad \mathbf{P} = \begin{bmatrix} 1.0, & 0.0, & 0.0 \\ 0.0, & -1.0, & 0.0 \\ 0.0, & 0.0, & -1.0 \end{bmatrix}$$

where $\mathbf{x}_2$ is the vector having the elements x_2, y_2, z_2. Fractional coordinates for statistically coexisting sheet structure are generated by the 2-fold rotation axis parallel to the b-axis located at $x = 0.125$ and $z = 0.375$.

Table 6
Observed and calculated structure factors

hkl[a]	$\sqrt{I_O}$	$\sqrt{I_C}$[b]	*hkl*[a]	$\sqrt{I_O}$	$\sqrt{I_C}$[b]	*hkl*[a]	$\sqrt{I_O}$	$\sqrt{I_C}$[b]
100, 010	14.9	14.8	331	–	7.9	603, 613, 623, 543, 453	22.4	23.0
			421, 241	8.6	9.2			
110	–	36.8						
			501, 511, 431, 341, 151, 051	–	12.6	263, 163, 063		
200, 020	98.7	98.2						
210, 120	122.8	121.8				703, 643, 723, 713, 553, 073, 173, 273, 463	20.5	21.8
220	–	8.3						
			102, 012	17.2	12.8			
300, 310, 130, 030	53.2	51.4						
			112	–	13.2			
			202, 212, 122, 022	35.6	34.3			
320, 230	–	12.1				733, 373	–	6.0
400, 410, 140, 040	24.8	25.9	222	–	16.8			
						653, 563	–	0.7
			302, 312, 132, 032	23.5	24.1			
330	–	3.8				803, 743, 823, 813, 083, 183, 283, 473	14.2	16.0
420, 240	7.7	7.9	322, 232	–	17.6			
500, 510, 430, 340, 150, 050	–	19.1	402, 412, 142, 042	11.0	11.4			
						104, 014	11.6	8.8
			332	–	9.2			
						114	–	12.1
520, 250	–	4.4	502, 512, 432, 342, 152, 052	–	13.3			
						204, 024	–	5.1
440	–	6.9						
						214, 124	11.6	12.1
530, 350	–	5.9						
			522, 252	7.6	4.9	224	–	5.8
600, 610, 620, 540, 450, 260, 160, 060	35.2	33.4						
						304, 314, 134, 034	17.1	16.9
			442	–	5.2			
			532, 352	–	5.4			
						324, 234	–	4.9
			602, 612	10.6	11.8			
630, 360	–	2.5	162, 062			404, 414, 334	7.3	9.6
700, 710, 550	–	5.2	622, 542, 452	–	3.2	144, 044		

Table 6 (*Continued*)

hkl[a]	$\sqrt{I_O}$	$\sqrt{I_C}$[b]	*hkl*[a]	$\sqrt{I_O}$	$\sqrt{I_C}$[b]	*hkl*[a]	$\sqrt{I_O}$	$\sqrt{I_C}$[b]
170, 070			262			424, 244	–	2.2
			103, 013	22.8	23.3			
720, 640, 460, 270	–	4.5				504, 514, 434, 344, 154, 054	9.9	4.6
			113	17.0	16.7			
			203, 213, 123, 023	46.3	45.9			
730, 370	–	1.3						
						524, 254	–	3.4
650, 560	–	0.3						
			223	–	15.1			
						444	–	0.4
800, 810, 820, 740, 470, 280, 180, 080	17.3	14.3	303, 313, 133, 033	13.7	18.6			
						534, 354	–	3.8
			323, 233	–	14.1	604, 614, 624, 544, 454, 264, 164, 064	10.3	3.9
			403, 413, 143, 043	7.9	10.6			
101, 011	6.9	7.0						
111	–	25.6						
			333	–	5.8	634, 364	–	1.1
201, 021	29.7	35.6						
			423	–	4.6			
211, 121	70.1	71.0	243			105, 015	13.6	8.1
			503, 513, 433	–	12.0			
221	–	15.9				115	–	5.6
301, 311, 131, 031	–	41.3	343, 153, 053			205, 215, 125, 025	7.4	8.2
			523	–	4.2			
321, 231	–	26.7	253			225	–	3.9
			443	–	2.1			
401, 411, 141, 041	25.1	26.7						
			533, 353	–	5.2			

[a] *hkl* denotes the overlapped reflection of hkl and $\bar{h}$kl.
[b] $\sqrt{I_C}$ is calculated by the relation $\sqrt{\Sigma m_j F_j^2}$, where m_j and F_j are the multiplicity and structure factor of the jth reflection, respectively.

oriented sample and that the temperature parameter B_y is far larger than B_x and B_z.

3.7. Diffuse streak scatterings along layer lines

Very weak streaks are observed between the equator and the first layer line, whose period corresponds to the sequence (-GAGAGS-)$_n$, three times as long as the sequence (-GA-)$_n$. The diffuse scattering intensity is given by the following equation,

$$I_{\text{diffuse}} = \langle |F|^2 \rangle - |\langle F \rangle|^2 \quad (1)$$

where $\langle\ \rangle$ indicates an average. If the serine residue is randomly substituted for an alanine residue, then Eq. (1) is approximately given as follows,

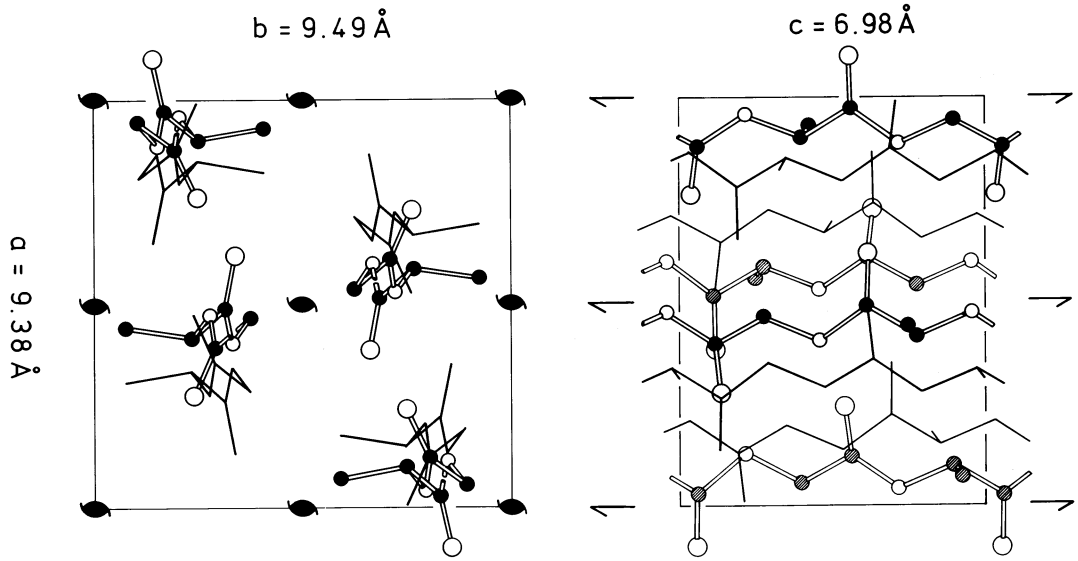


Fig. 6. The crystal structure of silk.

$$I_{\text{diffuse}} = |F_{\text{GAGAGA}} + F_{\text{O}}|^2 - |F_{\text{GAGAGA}} + 1/3F_{\text{O}}(\exp - 2\pi il/3 + 1 + \exp 2\pi il/3)|^2$$
$$= \{1 - 1/3(1 + \cos 2\pi il/3)\}[|F_O|^2 \times \{1 + 1/3(1 + 2\cos 2\pi il/3)\} + F_{\text{GAGAGA}}F^*_{\text{O}} + F_{\text{GAGAGA}}{*}F_{\text{O}}] = 0 \quad \text{when } l = 3n$$
$$|F_{\text{O}}|^2 + F_{\text{GAGAGA}}F^*_{\text{O}} + F_{\text{GAGAGA}}{*}F_{\text{O}} \quad \text{when } l \neq 3n$$

where F_{GAGAGA} is the structure factor for the sequence -GAGAGA- and F_{O} is the structure factor for the O(H) atom of serine residue. Since F_{GAGAGA} does not contribute to the layer line with $l \neq 3n$, I_{diffuse} reduces to $|F_{\text{O}}|^2$ on the layer lines with $l \neq 3n$. This suggests that the contribution of serine residue appears on the layer lines with $l \neq 3n$ in the period of $(\text{-GAGAGS-})_n$. The observed intensity distributions on the layer lines with $l \neq 3n$ monotonically decrease and correspond well with $|F_{\text{O}}|^2$.

The diffuse streak scatterings on the layer lines with $l = 3n$, which corresponds to the layer lines of the sequence $(\text{-GA-})_n$ should be attributed to the disorder in stacking of the sheet structure formed by hydrogen bonds.

$$I_{\text{diffuse}} = W_{BC}|F_{BC}|^2 + W_{B\bar{C}}|F_{B\bar{C}}|^2 - |W_{BC}F_{BC} + W_{B\bar{C}}F_{B\bar{C}}|^2$$
$$= W_{BC}W_{B\bar{C}}|F_{BC} - F_{B\bar{C}}|^2$$

where F_{BC} and $F_{B\bar{C}}$ are the structure factors of the sheet structures with the orientations BC and $B\bar{C}$. Furthermore, there may exist the same disorder, glide, as found in Nylon 6 and 66, where the sheet structures formed by hydrogen bonds slip relative to each other [28–30]. In this type of disorder, the regularity in a- and c-directions is kept, therefore, the diffuse streak scatterings should be observed on the lines parallel to the b*-direction.

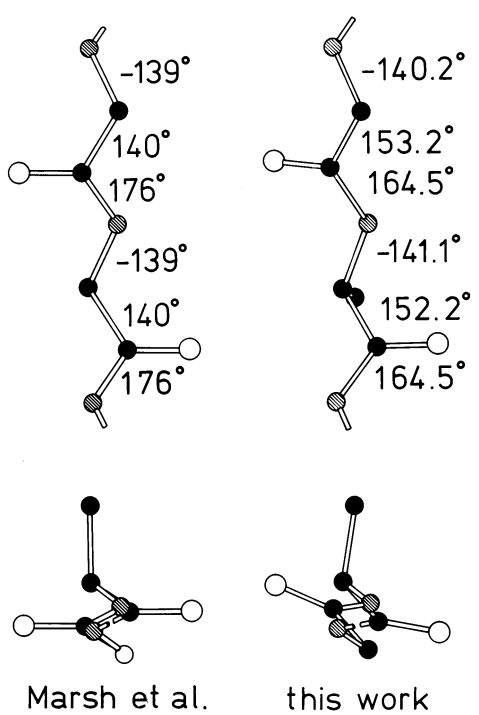


Fig. 7. The molecular structure of silk and the model proposed by Marsh et al. [6].

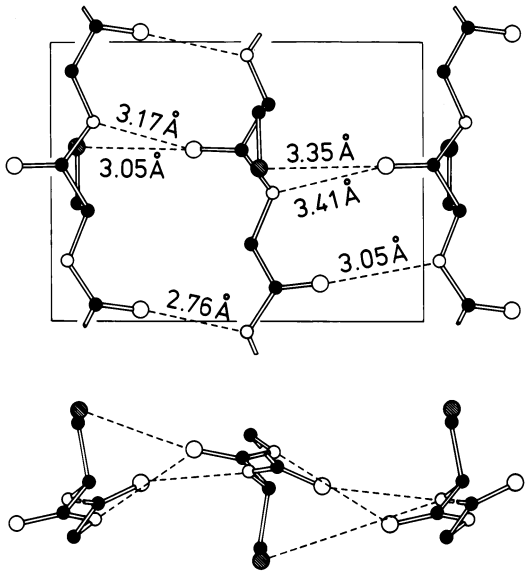


Fig. 8. The hydrogen bonding network of silk. The hatched circles indicate the O(H) atom of serine residue.

4. Conclusion

Reexamination of the crystal structure of silk (*Bombyx mori*) was successfully carried out by X-ray diffraction method. Four molecular chains are contained in the rectangular unit cell with parameters, $a = 9.38$ Å, $b = 9.49$ Å and c (fiber axis) $= 6.98$ Å, and the space group $P2_1-C_2^2$. Silk assumes the statistical crystal structure, in which two antipolar–antiparal lel sheet structures with different orientations statistically occupy a crystal site with the ratio 2:1. The molecular conformation is essentially the same pleated sheet structure as Marsh, Corey and Pauling proposed [6]. However, the sheet structure formed by hydrogen bonds assumes the antipolar–antiparallel structure different from that proposed by Marsh, Core and Pauling [6], in which the methyl groups of alanine residues alternately point to both sides of the sheet structure along the hydrogen bonding direction. This sheet structure is stereochemically more reasonable than that proposed by Marsh et al. [6]. The crystalline region of silk is composed of stacking of two antipolar–antiparallel sheet structures with different orientations. The diffuse streak scatterings observed between the layer lines can be interpreted by the existence of serine residues, i.e. the sequence of $(\text{-GAGAGS-})_n$.

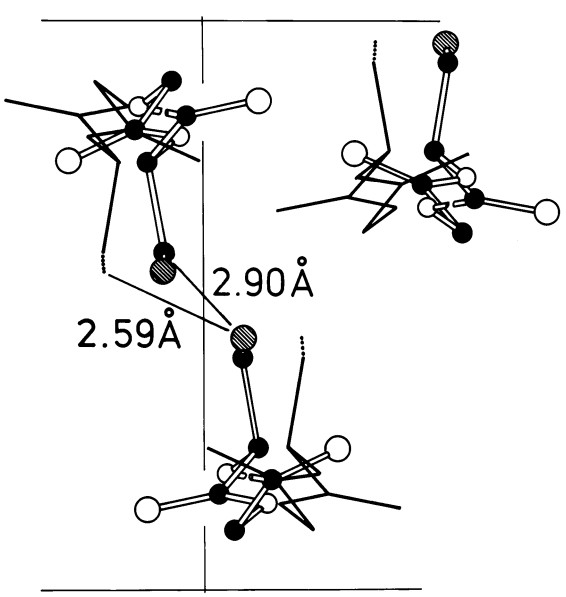


Fig. 9. Intermolecular contacts shorter than sums of the van der Waals radii.

Appendix A. Introduction of pseudosymmetries and existence probabilities to the constrained least-squares method

In the constrained least-squares method [14,15], the internal coordinates (bond lengths, bond angles and internal rotation angles) are used as variable parameters, and therefore, the bond lengths and bond angles can be easily fixed to the accepted values. At the present, the constrained least-squares refinement is a usual procedure for the structure analyses of the fibrous substances.

The constrained least-squares method is the one to minimize the following equation,

$$\Omega = \sum w_m \, \Delta F_m^2 + \sum \lambda_h G_h \tag{A1}$$

where ΔF_m is the difference between the observed and calculated structure factors of the mth reflection, w_m is the weight of the mth reflections, G_h is the hth constraining condition and λ_h is the undetermined multiplier. Usually, the undetermined multipliers are used for constraining the distances between asymmetric units in a molecule [15]. The structure factor having the index **h** is expressed by the following equation,

$$F(h) = \sum \sum f_j \exp[2\pi i \mathbf{h}(\mathbf{S}_s \mathbf{x}_j + \mathbf{T}_s)] \tag{A2}$$

$$\mathbf{x}_i = \mathbf{C}\mathbf{X}_i^c + \mathbf{T} = \mathbf{C}\mathbf{R}\mathbf{X}_i^m + \mathbf{T} \tag{A3}$$

$$\mathbf{X}_i^m = \mathbf{B}_{12} + \mathbf{A}_{12}\mathbf{B}_{23} + \ldots + \mathbf{A}_{12}\mathbf{A}_{23}\ldots\mathbf{A}_{i-2,\,i-1}\mathbf{B}_{i-1,\,i} \tag{A4}$$

$$\boldsymbol{A}_{ij} = \begin{bmatrix} -\cos\phi_j, & -\sin\phi_j, & 0 \\ \sin\phi_j \cos\tau_{ij}, & -\cos\phi_j \cos\tau_{ij}, & -\sin\tau_{ij} \\ \sin\phi_j \sin\tau_{ij}, & -\cos\phi_j \sin\tau_{ij}, & \cos\tau_{ij} \end{bmatrix}$$

$$\mathbf{B}_{ij} = \begin{bmatrix} b_{ij} \\ 0 \\ 0 \end{bmatrix}$$

where f_j is the atomic scattering factor of the jth atom, $\mathbf{S}_s$ and $\mathbf{T}_s$ are the symmetry operation matrix and vector of the space group, and $\mathbf{x}_j$ is the fractional coordinates, $\mathbf{X}_i^c$ is the Cartesian coordinates parallel to the crystal system, $\mathbf{X}_i^m$ is the Cartesian coordinates fixed on the molecule, $\mathbf{T}$ is the translational vector in the fractional coordinates, $\mathbf{C}$ is the transformation matrix from the Cartesian coordinates to the fractional coordinates, $\mathbf{R}$ is the transformation matrix including Euler angles, and b_{ij}, ϕ_j, and τ_{ij} are the bond length, bond angle, and internal rotation angle.

In the constrained least-squares method used in the present study, the structure factor is given by the following equation,

$$F(h) = \sum\sum W_M \sum W_M \sum\sum W_i f_i$$

$$\times \exp\{2\pi i \mathbf{h}[\mathbf{S}_s(\mathbf{S}_M^{inter}\mathbf{x}_{mji} + \mathbf{T}_M^{inter}) + \mathbf{T}_S]\} \quad (A5)$$

$$\mathbf{x}_{mji} = \mathbf{C}\mathbf{X}_{mji} + \mathbf{x}_m^O \quad (A6)$$

$$\mathbf{X}_{mji} = \mathbf{\Phi}_m \mathbf{P}_m \mathbf{X}_{ji} \quad (A7)$$

$$\mathbf{X}_{ji} = \mathbf{S}_j^{intra}\mathbf{X}_i + \mathbf{T}_j^{intra} \quad (A8)$$

$$\mathbf{X}_i = \mathbf{R}\mathbf{X}_i^A + \mathbf{X}^O \quad (A9)$$

where $\mathbf{S}_M^{inter}$, $\mathbf{T}_M^{inter}$, and W_M are the operation matrix, vector, and probability for the intermolecular pseudosymmetry, $\mathbf{\Phi}_m$ is the matrix including azimuthal angle of the molecule, $\mathbf{P}_m$ is the matrix denoting the direction and handedness of the molecule, W_m is the probability of the molecule, $\mathbf{S}_j^{intra}$ and $\mathbf{T}_j^{intra}$ are the operation matrix and vector for the intramolecular pseudosymmetry, $\mathbf{x}_m^O$ and $\mathbf{X}^O$ are the translational vectors in the fractional and Cartesian coordinates, and $\mathbf{X}_i^A$ is the Cartesian coordinates of the atom in an asymmetric unit represented by Eq. (A4). The probabilities, W_M and W_m, are constrained by the following equations,

$$\sum W_M = 1$$

$$\sum W_m = 1$$

This constrained least-squares method was applied to the structure refinements of Poly(vinylidene fluoride) form II [24], and silk.

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